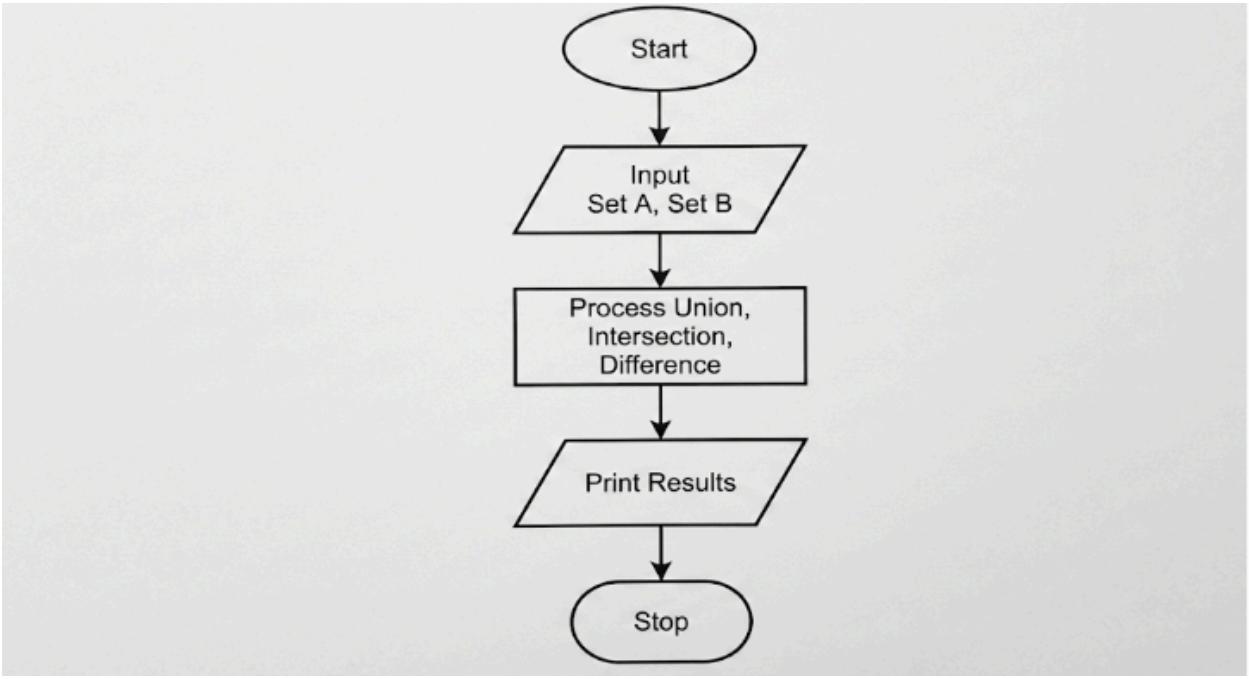
Experiment 4.1 Set Operations

Algorithm:

```
Step 1: START
Step 2: INPUT Set A — prompt "Set A: ", read space-separated integers, store
as set
Step 3: INPUT Set B — prompt "Set B: ", read space-separated integers, store
as set
Step 4: CALCULATE  union        = Set A | Set B
Step 5: CALCULATE  intersection = Set A & Set B
Step 6: CALCULATE  difference   = Set A - Set B
Step 7: OUTPUT "Union: "        followed by union result
Step 8: OUTPUT "Intersection: " followed by intersection result
Step 9: OUTPUT "Difference: "   followed by difference result
Step 10: STOP
```

Flowchart:



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4.1.1. Set Operations

06:01

Write a Python program to perform union, intersection and difference operations on $Set\ A$ and $Set\ B$.

Input Format:

- First Line prompts "Set A: " followed by space-separated list of integers for $Set\ A$.
- The second input prompts "Set B: " followed by space-separated list of integers for $Set\ B$.

Output Format:

- The first line prints "Union: " followed by the union of $Set\ A$ and $Set\ B$.
- The second line prints "Intersection: " followed by the intersection of $Set\ A$ and $Set\ B$.
- The third line prints "Difference: " followed by the difference of $Set\ A$ and $Set\ B$.

Note:

- If there is no intersection between the two sets, the program prints an empty set, which appears as "set()" in the output.
- Please refer to the visible test cases for better understanding.

Sample Test Cases +

Explorer

setoperat... Submit

```
set_a = set(map(int,input("Set A: ").split()))
set_b = set(map(int, input("Set B: ").split()))
print("Union:", set_a | set_b)
print("Intersection:",set_a & set_b)
print("Difference:", set_a - set_b)
```

Debugger

Average time	Maximum time
0.029 s	0.033 s
29.50 ms	33.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 27 ms Debug

Expected output	Actual output
Set A: 0 2 4 5 8	Set A: 0 2 4 5 8
Set B: 1 2 3 4 5	Set B: 1 2 3 4 5
Union: {0, 1, 2, 3, 4, 5, 8}	Union: {0, 1, 2, 3, 4, 5, 8}
Intersection: {2, 4, 5}	Intersection: {2, 4, 5}
Difference: {0, 8}	Difference: {0, 8}

Terminal Test cases

Prev Reset Submit Next

Code